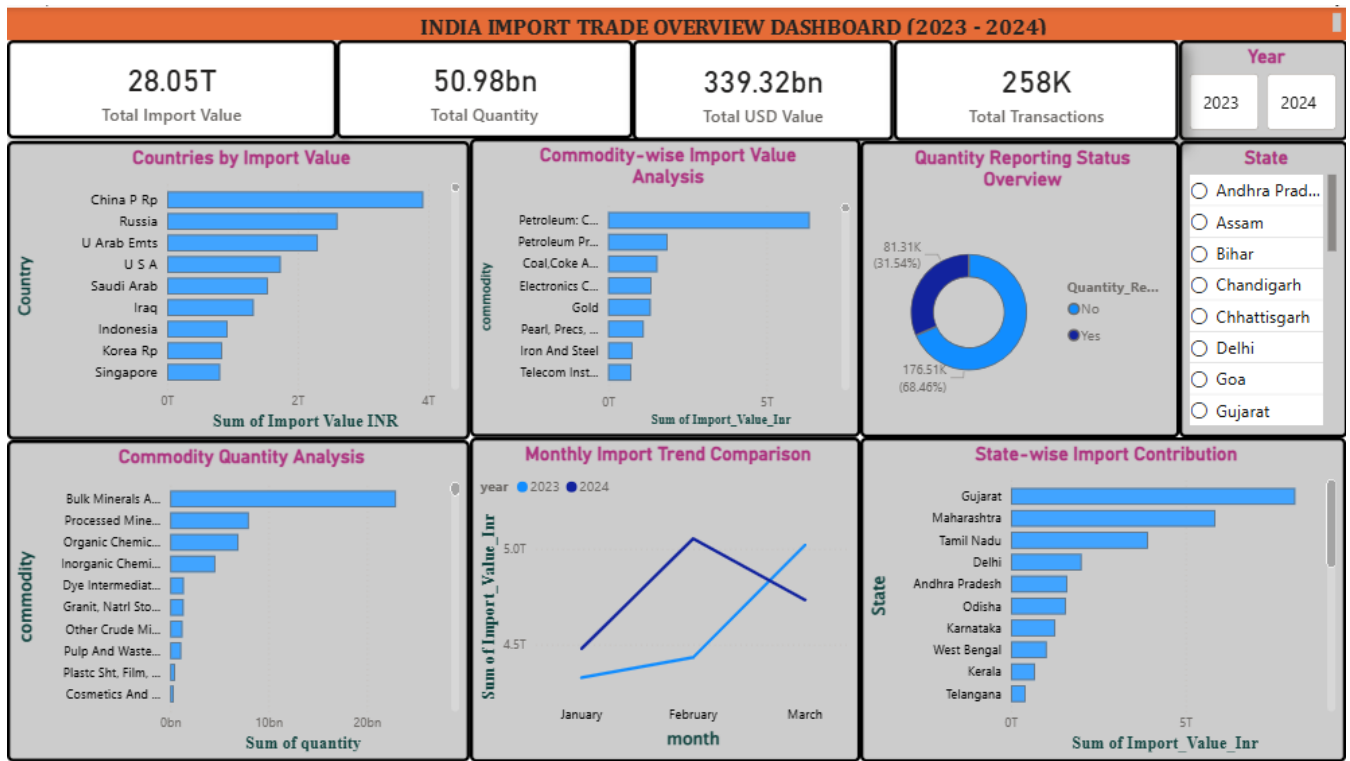
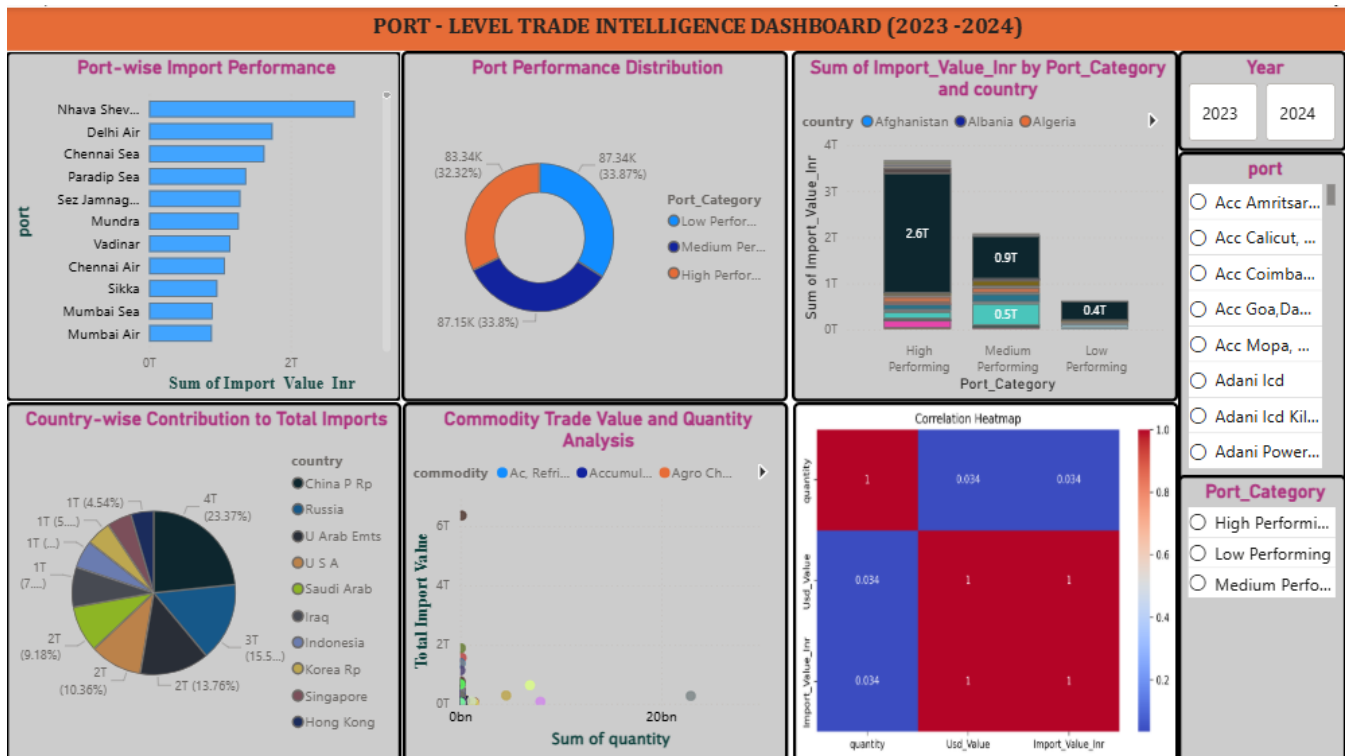


Port-Level Import Analytics and Trade Intelligence in India

India Import Trade Overview Dashboard



Port-level Trade Intelligence Dashboard



Dashboard Overview

Trade Performance Analysis: Monitors key import metrics, including import value, USD trade value, import quantity, and overall trade activity across the study period.

Import Pattern Analysis: Evaluates country-wise, commodity-wise, and state-wise import contributions to identify major trading partners, imported commodities, and regional trade distribution.

Trade Intelligence and Port Analysis: Assesses port performance, trade concentration, commodity value patterns, and import trends to support data-driven trade intelligence and decision-making.

Key Insights

- The dataset reveals that India's imports are concentrated among a few major countries, commodities, ports, and states. It highlights trade dependency patterns, import value distribution, and the key factors driving India's import ecosystem during 2023–2024
- India's import trade during 2023–2024 was highly concentrated among a few major trading partners, with China, Russia, and the UAE contributing the largest share of import value.
- **Petroleum-related** commodities dominated India's imports, indicating a strong dependence on energy and fuel imports to support industrial and economic activities.
- **Gujarat, Maharashtra, and Tamil Nadu** emerged as the leading import-contributing states, highlighting the importance of coastal and industrial regions in India's trade network.
- Major ports such as **Nhava Sheva, Chennai Sea, and Delhi Air** handled a significant portion of import trade, making them key gateways for international commerce.
- The analysis revealed that commodities with the highest import quantities were not necessarily the highest contributors to import value, indicating that commodity pricing plays a crucial role in determining trade value.
- Correlation analysis showed a strong relationship between **USD trade value** and **Import Value** (INR), while quantity had only a weak relationship with import value, confirming that trade expenditure is influenced more by commodity value than by volume.
- Import trends show increased activity during **2024**, suggesting growing domestic demand, industrial expansion, and higher trade engagement.
- The findings highlight the importance of trade diversification, efficient port infrastructure, and strategic sourcing to ensure a resilient and sustainable import ecosystem for India.

Recommendations

Country Analysis

- China, Russia, and UAE contribute a large share of imports.
- India should import from **more countries** instead of depending heavily on only a **few countries**. If one country stops supplying goods.
- India can easily buy from another country and trade will continue without major disruption.

Commodity Analysis

- Petroleum Crude contributes the highest import value (₹6.4 Trillion).
- India should gradually invest more in **renewable energy** such as solar and wind power and reduce dependence on imported fuel.
- This can help reduce import costs, improve energy security, and reduce the impact of global oil price fluctuations on the Indian economy.

Port Analysis

- Nhava Sheva and Chennai Sea handle a large share of India's imports.
- India should improve and **expand major ports** and develop other ports as well.
- Imports can move faster, congestion can be reduced, and future trade growth can be handled more efficiently.

Correlation Analysis

- Correlation analysis showed a strong relationship between USD Value and Import Value (INR).
- When the USD value of imports increases, India's import expenditure also increases.
- India should closely **monitor global commodity prices** because if the USD value of imported goods increases, India's total import bill will also increase.

Conclusion

- The Port-Level Import Analytics and Trade Intelligence in India (2023–2024) project successfully transformed raw import trade data into meaningful business insights through data cleaning, exploratory data analysis, visualization, and interactive dashboards.
- The analysis identified key trading partners, high-value imported commodities, major import-contributing states, and strategic ports driving India's import ecosystem.
- Overall, the project demonstrates how data analytics can be used to monitor trade performance, uncover import patterns, and generate actionable trade intelligence, enabling more informed decision-making for trade planning, infrastructure development, and long-term economic sustainability.